ResInsight: AI-Driven Developer Ecosystem

KUNJAL AGRAWAL

Master of Science in Computer Science

UC DAVIS

Advisor:- Prof. Mohammad Sadoghi

UCDAVIS

ExpoLab Creativity Unfolded

Apache ResilientDB incubating

Problem Statement

- Today's business applications are complex in nature with **interleaved tech stacks** with a mix of libs & frameworks, **code bases** are **evolving** rapidly on composable & interoperable stacks.
- E.g. code delivered 3-months back might look very different depending on the issues resolved or features built or new tech stack inclusion.
- **Why not AI Agents?** General-purpose AI tools **lack context** about your specific codebase. They're **trained** on **public data** but have no connection to your actual repositories, your project's conventions, or your **domain-specific implementations**

The Solution: ResInsight

ResInsight bridges this gap by connecting AI agents directly to your GitHub repositories through the Model Context Protocol (MCP).

Instead of asking ChatGPT general questions and getting generic answers, you can now ask questions like:

- “Show me all files related to transaction processing in ResilientDB”
- “How does the PBFT consensus implementation connect to the executor layer?”
- “What are the dependencies for the Debitable application?”

What Makes ResInsight Different?

1. **Repository-Aware Intelligence**
2. **Hybrid Search Architecture**
3. **Domain-Specific ResilientDB Knowledge**
4. **Private Repository Access**
5. **Tool-First Design**

Scenario: Exploring ResilientDB Applications

The Traditional Experience:

When assigned to work with a ResilientDB application like Debitable or Arrayán:

- Clone the repository and hope the README is current
- Try to understand dependencies by reading package.json or requirements.txt
- Look for example code or tutorials (often finding none)
- Attempt setup following possibly outdated instructions
- Debug environment issues specific to that application
- Eventually piece together understanding through experimentation

This exploration phase could consume significant time that should have been spent on project development.

Scenario: Exploring ResilientDB Applications

With ResInsight:

“What is Debitable and how does it integrate with ResilientDB?”

- Retrieves curated overview from the ResilientDB knowledge base and shows file structure
- Explains Debitable’s purpose, architecture, and key features
- Provides setup guidance specific to Debitable’s requirements
- Answers follow-ups about GraphQL integration, data models, or specific features

Then ask: *“Show me how Debitable handles data uploads”*

- Uses semantic search to find relevant React components
- Identifies files like DataUploader.jsx, InventoryPage.jsx
- Shows how these connect to ResilientDB API calls

Result: From zero understanding to basic comprehension in 15-20 minutes, with confidence to start building

Technical Architecture

- **Model Context Protocol (MCP)**
- **Search and Retrieval System**
  a. **Vector Embeddings (FAISS)**
  b. **Repository Indexing**:
  c. **Knowledge Graphs (NetworkX)**:

Security

Security is built into every layer:

- **MCP Token Authentication**: All client requests require valid tokens
- **GitHub PAT Authorization**: Read-only repository access
- **Environment-Based Secrets**: No credentials in code
- **Request Validation**: All inputs sanitized and validated

Impact

- **Time Savings**: Tasks that took days now take hours. Questions that required waiting for senior developers get answered immediately.
- **Independence**: New developers can explore and learn at their own pace, without constantly interrupting others.
- **Confidence**: When setup actually works the first time, when you understand why code is structured a certain way, when you can find what you need, that confidence compounds over time.
- **Knowledge Access**: Information that was previously locked in senior developers' experience is now available to everyone

Conclusion

- ResInsight was born from a genuine need I experienced as a student learning distributed systems and blockchain technology.
- It’s not trying to replace the excellent work being done by the ResilientDB team, the growing documentation efforts, or the collaborative spirit of the ExpoLab community.
- Instead, ResInsight addresses a specific gap: the mechanical obstacles that slow down learning.

THANK YOU